Tutorial Week 2

.

Answer the following permutation problems.

1. How many binary strings (strings made up of just 0s and 1s) of length 10 are there?
2. How many binary strings of length 10:
   (a) start and end with 1s.
   (b) exactly 2 ones? can you figure out how to rephrase this as a permutation question?
3. How many permutations of the letters ABCDEFGH contain
   (a) ED
   (b) CDE
   (c) BA and FGH
4. How many 4 digit numbers are odd?
5. How many ways can one win a dice game played with three distinguishable dice in which a winning roll is one that has at least two values that are the same. HINT: Try colouring the dice so you have red, green and black. Now how can you phrase this as a permutation?
6. How many positive integers less than 1,000 consist of distinct digits from $\{1, 3, 7, 9\}$? Hint: the sum rule is involved.

   (a) How many arrangements are there of the letters in the word *EMPHATIC*?
   (b) Of these, how many have the letters *E* and *M* side by side in either *EM* **or** *ME*?